SignVaani: Project Progress Report

Real-Time Indian Sign Language to Speech, B.Tech CSE mini-project, progress up to 2 November 2026

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1. Project at a glance

Item	Details
Problem	Deaf patients who use Indian Sign Language (ISL) cannot easily talk to hospital staff who do not know ISL, and certified interpreters are scarce
Solution	Laptop webcam -> MediaPipe body, hand and face landmarks -> sign recognition -> grammatical sentence -> English or Hindi speech, on a CPU, without a GPU
Vocabulary (frozen)	15 hospital signs + IDLE: ME, DOCTOR, PAIN, FEVER, HEAD, STOMACH, WATER, MEDICINE, TOILET, HELP, NEED, CALL, WHERE, YES, NO
Demo sentences (frozen)	S1-S10, e.g. ME STOMACH PAIN -> "My stomach hurts." / "मेरे पेट में दर्द है।"
Main model	BiGRU with attention pooling (about 0.69 M parameters), compared with a Transformer and a 1D-CNN
Evaluation plan	Signer-independent: leave-one-signer-out over 5 signers, plus a 6th person never seen in training
Technology	Python 3.10/3.11, MediaPipe 0.10.14, OpenCV, PyTorch (CPU), ONNX Runtime, scikit-learn, pyttsx3, gTTS
Work so far	23 October to 2 November 2026, 15 commits on GitHub

Summary of status

- **Software: implemented.** Every module needed for recording, quality control, training, evaluation, model export, the live app, sentence generation and speech is written, with 67 automated unit tests.
- **Design and documents: complete.** SRS, four design diagrams, test plan with 18 test cases, literature survey of 24 papers, IEEE paper draft, recording protocol and consent form, PPT content.
- **Pending:** recording the team's own dataset, training runs and their measured results, the live test cases, the final report and the demo video.

2. Objectives and their status

#	Objective	Implemented	Measured	Evidence / next step
1	Recognise 15 signs + IDLE from a webcam in real time (at least 15 FPS) on a CPU	Done	To do	`app.py`, `models.py`, ONNX export; FPS and accuracy measured after training (TC-01, TC-05)
2	Detect the start and end of each sign automatically, ignore non-sign movement	Done	To do	`spotter.py` with motion gate + IDLE class; to be measured live (TC-09, TC-10)
3	Turn ISL word order into grammatical English and Hindi sentences and speak them	Done	Verified	All 11 demo sentences correct in both languages (Section 6)
4	Evaluate honestly on unseen signers; never store video	Done	To do	`train.py --loso`, `evaluate.py`; recorder saves only landmarks (TC-18)

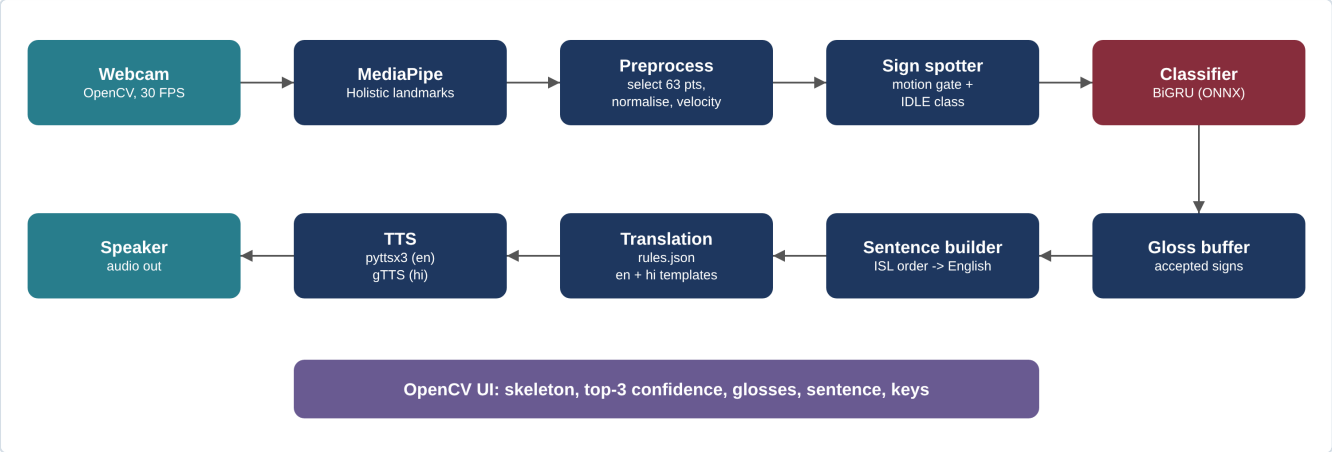
3. Work timeline

From the commit history on GitHub.

Date	Milestone	What was delivered
23 Oct	Project set-up	Python package skeleton; project blueprint (1-week and 8-week plans), literature survey with 24 papers, IEEE paper draft, dataset guide
23 Oct	Public dataset	Tool to download the INCLUDE ISL dataset (AI4Bharat, 263 signs, 4,292 videos) and convert the videos to keypoints; official split lists
23 Oct	Day 1: scope and set-up	Vocabulary and sentences frozen (and locked by tests); install check; recorded-data check; guided recorder that resumes and never overwrites; SRS; architecture and DFD level 0 diagrams; recording protocol, consent form, sign verification sheet
26 Oct	Days 2-3: data and baseline tools	Dataset statistics (Table R1); quality control (reject, review, re-record list); unit tests for S1-S10; confusion-matrix analysis (Tables R3, R6); use-case and sequence diagrams; datasheet; INCLUDE-50 real-data check
29 Oct	App and training improvements	Camera preview mode before a model exists; resizable and full-screen window; class-balanced loss; training clips cut to the live spotter window; on-screen "added / not added" message
30 Oct	Day 4: models and speed	Model comparison (BiGRU, Transformer, 1D-CNN); ablation study; final model export to ONNX with a check that ONNX equals PyTorch; FPS and latency measurement; test case document TC-01 to TC-18
30 Oct	Day 5: sentences and speech	Sentence + speech end to end in English and Hindi; Hindi audio cache for offline use; unseen-signer (s06) evaluation; bug-bash plan; speech made one-at-a-time
2 Nov	Review preparation	Slide-by-slide PPT content (with PDF), diagram images, this progress report

4. What has been built

Figure: `docs/diagrams/architecture.png`. The top row runs on every frame; the bottom row runs once per accepted sign or sentence.



architecture.png

Modules (21 Python modules, about 2,900 lines of code)

Area	Modules	Lines
Data collection and checks	`record.py`, `check_install.py`, `check_data.py`, `qc.py`, `dataset_stats.py`	787
Public dataset (INCLUDE)	`include_dataset.py`	345
Features and augmentation	`landmarks.py`, `augment.py`, `data.py`	301
Models and training	`models.py`, `train.py`, `experiments.py`	463
Evaluation	`evaluate.py`, `confusion.py`	337
Deployment	`export_onnx.py`, `inference.py`	113
Real-time app	`app.py`, `spotter.py`	382
Language and speech	`sentence.py` + `rules.json`, `speech.py`	156
Test data generator	`make_synthetic.py` (for automated tests only)	51

Documents and design

Document	Content	Status
`docs/SRS.md`	12 functional and 7 non-functional requirements, each traced to a module and a test case; measurable targets; feasibility	Done
`docs/diagrams/`	Architecture, DFD level 0, use-case and sequence diagrams (draw.io, SVG and PNG)	Done
`docs/test_cases.md`	18 system test cases with how to run, expected result and linked unit test	Done

Document	Content	Status
`docs/recording_protocol.md`	Recording checklist, consent form, consent register	Done
`docs/sign_verification.md`	Sign references (ISLRTC dictionary), ISL-user and Hindi checks	Done
`data/DATASHEET.md`	Datasheet for our recordings and for INCLUDE	In progress
`docs/presentation_content.md`	Content for every slide of the review PPT	Done
Project report	Chapters 1-6	To do

Testing

- **67 automated unit tests** in 14 test files (1,246 lines) cover landmark processing, sign spotting, training and splits, sentences, speech, recorder, quality control, the app and the INCLUDE tool. Run `python -m pytest -q` on the demo laptop before the review and quote the pass count.
- **18 system test cases** (TC-01 to TC-18) are written and will be run with the final model.

5. Method implemented

- **Landmarks:** MediaPipe Holistic gives body, hand and face points; we keep 63 per frame (9 upper-body, 12 face, 21 per hand). Missing points are stored as NaN; gaps of up to 6 frames are filled.
- **Features:** each sign is resampled to 32 frames, centred on the shoulders and divided by shoulder width; velocity is added, giving a **32 x 378** input.
- **Augmentation:** speed 0.8-1.2x, rotation ±15°, scale 0.9-1.1, shift, noise, and hiding a hand for a few frames.
- **Model:** Linear 378 -> 256 + LayerNorm -> 2-layer bidirectional GRU (hidden 128) -> attention pooling -> 16 classes. Transformer and 1D-CNN versions use the same input for comparison.
- **Training:** AdamW (learning rate 0.001), cosine decay, label smoothing 0.1, class-balanced loss, up to 120 epochs with early stopping (patience 25), batch 64.
- **Evaluation:** random split (same people) and leave-one-signer-out (new people), top-1 accuracy, macro-F1, confusion matrices, and a final test on a 6th signer.
- **Live use:** a sign starts when the wrists move and ends after 8 still frames; it is accepted only if confidence is at least 0.8 and it is not IDLE. Accepted signs go to the sentence rules and are spoken (pyttsx3 for English offline, gTTS with a cache for Hindi).

6. Results so far

Item	Result	Status
Sentence generation, English (S1-S10, 11 sentences)	11 / 11 exactly as expected	Verified
Sentence generation, Hindi (S1-S10, 11 sentences)	11 / 11 exactly as expected	Verified

Item	Result	Status
Classifier size (BiGRU + attention)	694,801 parameters (about 0.69 M), small enough for a CPU	Verified
Frozen scope	Vocabulary and sentences locked by `tests/test_frozen_scope.py`	Done
Accuracy, random split and leave-one-signer-out	[XX] % and [XX] ± [XX] %	To do
Frame rate, latency, false triggers while idle	[XX] FPS, [XX] ms, [XX] per minute	To do
Unseen signer s06 (60 attempts)	[XX] / 60	To do

Accuracy and speed numbers are filled in only from the team's own runs after training.

7. Work remaining

Task	Status	Command / note
Record 5 signers x 16 classes x 10 takes (800 clips) and run quality control	To do	`python -m signvaani.record --signer s01`, then `python -m signvaani.qc --apply`
Extra experiment on public data: INCLUDE "Adjectives" part 4 of 8	In progress	Downloaded and unzipped; next: `include_dataset convert`, then `train` with the official split
Pipeline check on INCLUDE-50	To do	`python -m signvaani.include_dataset download --subset include50`
Training: random split, LOSO, model comparison, ablation	To do	`train --loso`, `experiments compare`, `experiments ablate`
Final model, ONNX export, live app tuning	To do	`train --final`, `export_onnx`, `app --model models/sign_model.onnx`
Record s06 (test only) and evaluate	To do	`evaluate --data data/unseen --signers s06`
Run system test cases TC-01 to TC-18	To do	On the weakest laptop, in the review room
Report, demo video, final PPT	In progress	PPT content ready; fill in measured numbers

8. Challenges and how we handled them

Challenge	How we handled it
MediaPipe installs only on certain Python versions	Pinned Python 3.10/3.11, MediaPipe 0.10.14 and NumPy below 2; an install-check script tests every laptop
The system could add words while the user is just resting	An IDLE class (resting hands, touching the face, adjusting hair), a motion gate, a 0.8 confidence threshold and a cool-down

Challenge	How we handled it
Training clips looked different from live windows	Training clips are cut to the same window the live spotter classifies
Signs with fewer clips were predicted less often	Class-balanced loss: rarer signs get a larger weight
Accuracy looks too high when the same person is in training and test	Leave-one-signer-out evaluation plus a sixth, unseen person
Hindi speech needs internet, which the review hall may not have	Hindi audio for S1-S10 is cached in advance; English works fully offline
Two quick SPACE presses started two speech engines	Sentences are now spoken one at a time

9. Likely review questions

Question	Short answer
Why landmarks instead of raw video?	Landmarks are small (378 numbers per frame), ignore the background and clothing, run fast on a CPU, and protect privacy because no video is stored
Why a BiGRU?	Our dataset is small; a 0.69 M-parameter recurrent model learns well from little data and runs in real time. We still compare it with a Transformer and a 1D-CNN
What is leave-one-signer-out?	Train on 4 people, test on the 5th, repeat for each person. It shows how the system works for a new patient
How does it know where a sign starts and ends?	Wrist speed above a threshold opens a window; 8 quiet frames close it; the IDLE class and the 0.8 confidence threshold reject non-signs
How is ISL grammar handled?	ISL order such as ME STOMACH PAIN is matched to patterns like [PRONOUN, BODY, PAIN] and turned into "My stomach hurts." and the Hindi sentence
What if there is no internet?	Recognition and English speech are fully offline; Hindi audio is cached beforehand
What are the limitations?	Isolated signs from one domain, a small number of signers, no full ISL grammar or fingerspelling. The tool supports interpreters; it does not replace them